Experiment:21

Code:

```
hanoi(1, Source, Destination, _) :-
   write('Move disk from '),
   write(Source),
   write(' to '),
   write(Destination),
   nl.


hanoi(N, Source, Destination, Auxiliary) :-
   N > 1,
   M is N - 1,
   hanoi(M, Source, Auxiliary, Destination),
   hanoi(1, Source, Destination, Auxiliary),
   hanoi(M, Auxiliary, Destination, Source).
```

Output:

```
32 ?- consult(['c:/Users/Dhar/OneDrive/Desktop/towers.p1']).
true.

33 ?- hanoi(3, a, c, b).
Move disk from a to c
Move disk from a to b
Move disk from c to b
Move disk from a to c
Move disk from b to a
Move disk from b to c
Move disk from a to c
true
```

Experiment:22

Code:

```
bird(parrot).

bird(pigeon).

bird(ostrich).

bird(penguin).


can_fly(sparrow).

can_fly(parrot).

can_fly(pigeon).


cannot_fly(ostrich).

cannot_fly(penguin).


check_flight(Bird) :-

  can_fly(Bird),

  write(Bird),

  write(' can fly.').


check_flight(Bird) :-

  cannot_fly(Bird),

  write(Bird),

  write(' cannot fly.').


check_flight(Bird) :-

  bird(Bird),

  \+ can_fly(Bird),

  \+ cannot_fly(Bird),
```

write('Flight information is unknown.').

Output:

```
34 ?- consult(['c:/Users/Dhar/OneDrive/Desktop/bird(sparrow).p1']).
true.

35 ?-  check_flight(sparrow).
sparrow can fly.
true .

36 ?- check_flight(penguin).
penguin cannot fly.
true .

37 ?- check_flight(ostrich).
ostrich cannot fly.
true
```

Experiment:23

Code:

male(john).

male(paul).

male(mike).

female(mary).

female(lisa).

female(anna).

parent(john, paul).

parent(mary, paul).

parent(john, lisa).

parent(mary, lisa).

```
parent(paul, mike).

parent(anna, mike).


father(X, Y) :-

   male(X),

   parent(X, Y).


mother(X, Y) :-

   female(X),

   parent(X, Y).


child(X, Y) :-

   parent(Y, X).


sibling(X, Y) :-

   parent(Z, X),

   parent(Z, Y),

   X \= Y.
```

Output:

```
38 ?- consult(['c:/Users/Dhar/OneDrive/Desktop/male.p1']).
true.

39 ?- father(john, paul).
true.

40 ?-  mother(mary, lisa).
true.

41 ?-  child(paul, john).
true.
```

Experiment:24

Code:

```
disease(diabetes).

disease(high_bp).

disease(obesity).


diet(diabetes, 'Low sugar and high fibre food').

diet(high_bp, 'Low salt and low fat food').

diet(obesity, 'Low calorie food and regular exercise').


suggest_diet(Disease) :-

  diet(Disease, Recommendation),

  write('Recommended diet: '),

  write(Recommendation).


suggest_diet(Disease) :-

  \+ diet(Disease, _),

  write('No diet recommendation available.').
```

Output:

```
49 ?- consult(['c:/Users/Dhar/OneDrive/Desktop/diabetes.pl']).
true.

50 ?- suggest_diet(diabetes).
Recommended diet: Low sugar and high fibre food
true .

51 ?- suggest_diet(high_bp).
Recommended diet: Low salt and low fat food
true .

52 ?- suggest_diet(obesity).
Recommended diet: Low calorie food and regular exercise
true
```

Experiment:25

Code:

```
can_get_banana :-
   monkey(on_floor),
   banana(on_ceiling),
   box(in_room),
   write('Monkey pushes the box under the banana.'), nl,
   write('Monkey climbs onto the box.'), nl,
   write('Monkey grabs the banana.'), nl,
   write('Monkey gets the banana!').


monkey(on_floor).
banana(on_ceiling).
box(in_room).
```

Output:

```
48 ?- consult(['c:/Users/Dhar/OneDrive/Desktop/banana.p1']).
true.

48 ?- can_get_banana.
Monkey pushes the box under the banana.
Monkey climbs onto the box.
Monkey grabs the banana.
Monkey gets the banana!
true.
```